

Technical Data

Product Description

ELCRES XD2322 polycarbonate (PC) copolymer resin is a UV stabilized opaque grade. This resin offers UL94 V0@1.5mm flame retardancy based on non-bromine, non-chlorine FR system, excellent low temperature ductility characteristics and processability. ELCRES XD2322 resin is an excellent candidate for a wide range of applications such as electronic vehicles (EV) and 5G devices.

General

Material Status	• Commercial: Active		
UL Yellow Card ¹	• E207780-104498338		
Search for UL Yellow Card	• SABIC • LNP™ ELCRES™		
Availability	• Africa & Middle East • Asia Pacific	• Europe • Latin America	• North America
Features	• Bromine Free • Chlorine Free	• Flame Retardant • High Impact Resistance	• UV Resistant
Uses	• Appliances • Decorative Parts • Electric Vehicle Applications • Electrical Parts	• Electrical/Electronic Applications • Electronic Displays • Lighting Applications • Medical Devices	• Optical Applications • Personal Care • Sporting Goods
Processing Method	• Injection Molding		

Physical	Nominal Value Unit	Test Method
Density / Specific Gravity	1.19 g/cm³	ASTM D792 ISO 1183
Melt Mass-Flow Rate (MFR) (300°C/1.2 kg)	16 g/10 min	ASTM D1238
Melt Volume-Flow Rate (MVR) (300°C/1.2 kg)	15 cm³/10min	ISO 1133
Molding Shrinkage		Internal Method
Across Flow : 3.20 mm	0.40 to 0.80 %	
Flow : 3.20 mm	0.40 to 0.80 %	
Water Absorption		ISO 62
Saturation, 23°C	0.40 %	
Equilibrium, 23°C, 50% RH	0.15 %	
Outdoor Suitability	f1	UL 746C
Mechanical	Nominal Value Unit	Test Method
Tensile Modulus		
-- ³	2100 MPa	ASTM D638
--	2000 MPa	ISO 527-1/1
Tensile Strength		
Yield ⁴	58.0 MPa	ASTM D638
Yield	56.0 MPa	ISO 527-2/50
Break ⁴	59.0 MPa	ASTM D638
Break	57.0 MPa	ISO 527-2/50
Tensile Elongation		
Yield ⁴	6.0 %	ASTM D638
Yield	6.0 %	ISO 527-2/50
Break ⁴	110 %	ASTM D638
Break	110 %	ISO 527-2/50
Flexural Modulus		
50.0 mm Span ⁵	2400 MPa	ASTM D790
-- ⁶	2100 MPa	ISO 178
Flexural Strength		
50.0 mm Span ⁵	92.0 MPa	ASTM D790
-- ⁶	87.0 MPa	ISO 178



Impact	Nominal Value Unit	Test Method
Notched Izod Impact		
-40°C	360 J/m	ASTM D256
-30°C	650 J/m	ASTM D256
23°C	780 J/m	ASTM D256
-30°C ⁷	29 kJ/m ²	ISO 180/1A
23°C ⁷	63 kJ/m ²	ISO 180/1A
Unnotched Izod Impact Strength ⁷		ISO 180/1U
-30°C	No Break	
23°C	No Break	
Instrumented Dart Impact		ASTM D3763
23°C, Total Energy	66.0 J	
Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load		
1.8 MPa, Unannealed, 3.20 mm	123 °C	ASTM D648
1.8 MPa, Unannealed, 4.00 mm, 64.0 mm Span ⁸	118 °C	ISO 75-2/Af
Vicat Softening Temperature	140 °C	ASTM D1525 ⁹
CLTE		ASTM E831
Flow : -40 to 40°C	6.1E-5 cm/cm/°C	
Transverse : -40 to 40°C	6.2E-5 cm/cm/°C	
RTI Elec	125 °C	UL 746B
RTI Imp	115 °C	UL 746B
RTI Str	125 °C	UL 746B
Electrical	Nominal Value Unit	Test Method
Comparative Tracking Index (CTI)	PLC 3	UL 746A
Comparative Tracking Index (Solution B)	150 V	IEC 60112
High Amp Arc Ignition (HAI)		UL 746A
1.5 mm	PLC 1	
3.0 mm	PLC 0	
Hot-wire Ignition (HWI)		UL 746A
1.5 mm	PLC 3	
3.0 mm	PLC 2	
Flammability	Nominal Value Unit	Test Method
Flame Rating		UL 94
> 1.5 mm	V-0	
> 3.0 mm	5VA	
Glow Wire Flammability Index (1.5 mm)	960 °C	IEC 60695-2-12
Glow Wire Ignition Temperature (1.5 mm)	825 °C	IEC 60695-2-13
Injection	Nominal Value Unit	
Drying Temperature	120 °C	
Drying Time	3.0 to 4.0 hr	
Suggested Max Moisture	0.020 %	
Suggested Shot Size	40 to 60 %	
Rear Temperature	270 to 295 °C	
Middle Temperature	280 to 305 °C	
Front Temperature	295 to 315 °C	
Nozzle Temperature	290 to 310 °C	
Processing (Melt) Temp	295 to 315 °C	
Mold Temperature	70 to 95 °C	
Back Pressure	0.300 to 0.700 MPa	
Screw Speed	40 to 70 rpm	
Vent Depth	0.025 to 0.076 mm	



Injection Notes

- Drying Time (Cumulative): 48 hr

Notes

¹ A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

² Typical properties: these are not to be construed as specifications.

³ 5.0 mm/min

⁴ Type I, 50 mm/min

⁵ 1.3 mm/min

⁶ 2.0 mm/min

⁷ 80*10*3 mm

⁸ 80*10*4 mm

⁹ Rate A (50°C/h), Loading 2 (50 N)



Where to Buy

Supplier**SABIC**

Web: <http://www.sabic.com/>

Distributor**Amco Polymers**

Telephone: 800-262-6685

Web: <http://www.amcopolymers.com/>

Availability: North America

Chase Plastic Services, Inc.

Chase Plastics Services is a North American distributor with representatives throughout the region. Please find your rep here: <http://www.chaseplastics.com/contact/locations>

Telephone: 800-232-4273

Web: <http://www.chaseplastics.com/>

Availability: North America

Nexeo Plastics

Nexeo Plastics is leading global resin distributor with the technical resources you need to overcome material challenges. Visit us on the web at www.nexeoplastics.com.

Telephone: 833-446-3936

Web: <https://www.nexeoplastics.com/>

Availability: North America

POLYMIX

POLYMIX is a Pan European distribution company. Contact POLYMIX for availability of individual products by country.

Telephone: +33-3-8920-1380

Web: <http://www.polymix.eu/>

Availability: France